Trigonometric Identities And Equations

Chapter Overview

Solution 2

Exact values

Derive the sine, cosine and tangent of 30°, 45° and 60°; use the unit circle to extend these ideas to other angles.

Trigonometric identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$
$$\sin^2\theta + \cos^2\theta = 1$$

Trigonometric equations

Solve equations such as $\sin(n\theta) = k$ and $\sin(\theta \pm \alpha) = k$ over a specified interval.

Quadratic equations

Solve equations that are quadratic in sine, cosine or tangent. Check the range of the function and the interval for the angle.

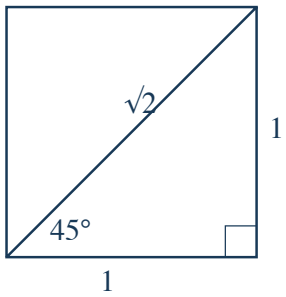
sin/cos/tan of 30°, 45°, 60°

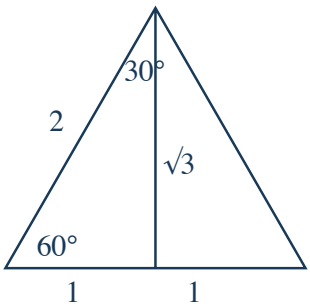
Solution 3

You will frequently encounter 30˚, 60˚ and 45˚ in geometric problems.
Why?

Why? We see these angles in equilateral triangles and half squares.

Although you have a calculator, you need to know how to derive these values.
Draw half a unit square and half an equilateral triangle of side 2.





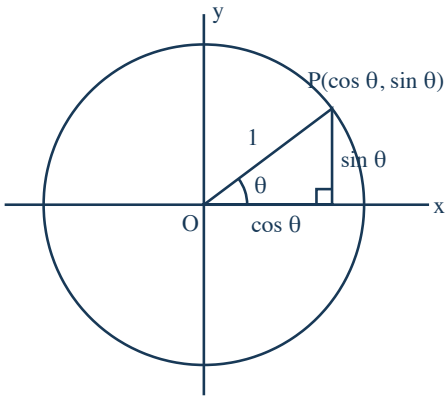
$\sin 45° =$	$1/\sqrt{2}$	$\cos 45° =$	$1/\sqrt{2}$	$\tan 45° =$	1
$\sin 30° =$	$1/2$	$\cos 30° =$	$\sqrt{3}/2$	$\tan 30° =$	$1/\sqrt{3}$
$\sin 60° =$	$\sqrt{3}/2$	$\cos 60° =$	$1/2$	$\tan 60° =$	$\sqrt{3}$

The Unit Circle and Trigonometry

Solution 4

The unit circle

A point P on the circle of radius 1 has coordinates $(\cos\theta, \sin\theta)$. Positive angles turn anticlockwise from the positive x-axis.



For an acute angle

$$\cos\theta = \frac{\text{adjacent}}{1}, \qquad \sin\theta = \frac{\text{opposite}}{1}$$

What is the gradient of OP?

Gradient

$$\text{gradient} = \frac{\text{rise}}{\text{run}} = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

The coordinates extend sine and cosine to angles in every quadrant. Tangent is undefined when cosine is zero.

Mini-Exercise

Solution 5

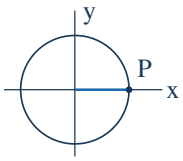
Use the unit circle to determine each value, using 0, +ve, −ve, 1, −1 or undefined.

Recall: $P = (\cos\theta, \sin\theta)$; $OP = 1$, and the gradient of OP is $\tan\theta$.

Cosine gives the x-value; sine gives the y-value.

Vertical lines do not have a well-defined gradient, so this value is undefined.

1. $\theta = 0°$

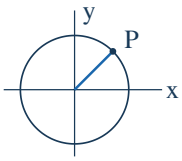


$\cos\theta = 1$

$\sin\theta = 0$

$\tan\theta = 0$

2. $0° < \theta < 90°$



2. values

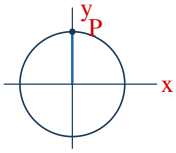
$\cos\theta =$ +ve

$\sin\theta =$ +ve

$\tan\theta =$ +ve

3. $\theta = 90°$

3. diagram



3. values

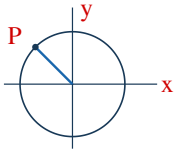
$\cos\theta = 0$

$\sin\theta = 1$

$\tan\theta =$ undefined

4. $90° < \theta < 180°$

4. diagram



4. values

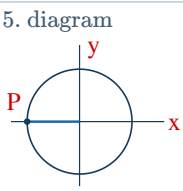
$\cos\theta =$ -ve

$\sin\theta =$ +ve

$\tan\theta =$ -ve

5. $\theta = 180°$

5. diagram



5. values

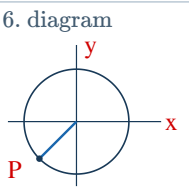
$\cos\theta = -1$

$\sin\theta = 0$

$\tan\theta = 0$

6. $180° < \theta < 270°$

6. diagram



6. values

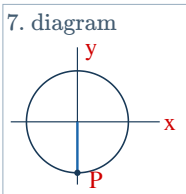
$\cos\theta =$ -ve

$\sin\theta =$ -ve

$\tan\theta =$ +ve

7. $\theta = 270°$

7. diagram



7. values

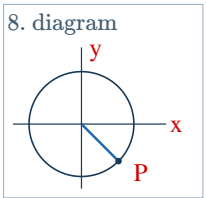
$\cos\theta = 0$

$\sin\theta = -1$

$\tan\theta =$ undefined

8. $270° < \theta < 360°$

8. diagram



8. values

$\cos\theta =$ +ve

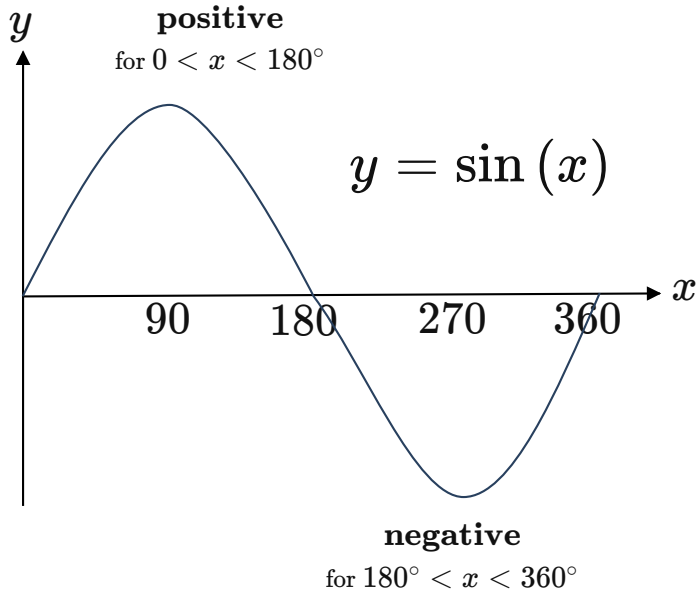
$\sin\theta =$ -ve

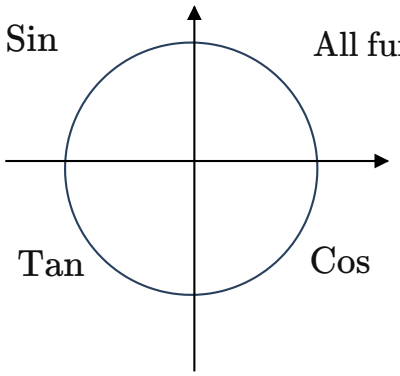
$\tan\theta =$ -ve

The Unit Circle and Trigonometry

Solution 6

The unit circles explains the behaviour of these trigonometric graphs beyond $90°$.
However, the easiest way to remember whether $\sin(x)$, $\cos(x)$, $\tan(x)$ are positive or negative is to just do a **very quick sketch (preferably mentally!)** of the corresponding graph.





Note: The textbook uses something called '*CAST diagrams*', which is an acronym for which quadrants the **C**osine, **A**ll, **S**ine and **T**angent are positive

A Few Trigonometric Angle Laws

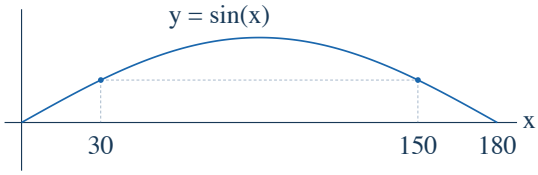
Solution 7

These laws follow from a quick sketch of a trigonometric graph. They are a convenience so you need not draw a graph every time. Memorising them helps you answer exam questions faster.

1. $\sin x = \sin(180° - x)$.

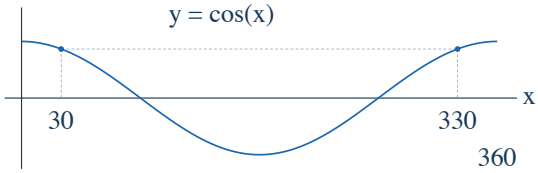
For example, $\sin 150° = \sin 30°$.

We saw this in the previous chapter: the ambiguous case when using the sine rule.



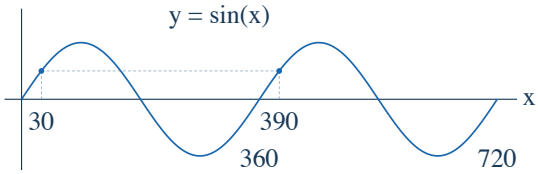
2. $\cos x = \cos(360° - x)$.

For example, $\cos 330° = \cos 30°$.



3. Sine and cosine repeat every 360°; tangent repeats every 180°.

For example, $\sin 390° = \sin 30°$.



4. $\sin x = \cos(90° - x)$.

For example, $\sin 50° = \cos 40°$.

Recall from the previous chapter: cosine is the sine of the complementary angle. This was not covered in the textbook, but appeared unexpectedly in a C3 exam.

Examples

Solution 8

Without a calculator, find each exact value.

(a) $\tan 225°$

a

$\tan 45° = 1$

(b) $\tan 210°$

b

$\tan 30° = \dfrac{1}{\sqrt{3}}$

(c) $\sin 150°$

c

$\sin 30° = \dfrac{1}{2}$

(d) $\cos 300°$

d

$\cos 60° = \dfrac{1}{2}$

(e) $\sin(-45°)$

e

$-\sin 45° = -\dfrac{1}{\sqrt{2}}$

(f) $\cos 750°$

f

$\cos 30° = \dfrac{\sqrt{3}}{2}$

(g) $\cos 120°$

g

$-\cos 60° = -\dfrac{1}{2}$

Sine and cosine repeat every 360°; tangent repeats every 180°. Use a reference angle and the sign in the quadrant.

Sine and tangent are odd functions: $f(-x) = -f(x)$. Cosine is even: $\cos(-x) = \cos x$.

Test Your Understanding — Solution 9

Without a calculator, work out the value of each below.

a. $\cos 315°$

a
$\cos 45° = \frac{1}{\sqrt{2}}$

b. $\sin 420°$

b
$\sin 60° = \frac{\sqrt{3}}{2}$

c. $\tan(-120°)$

c
$\tan 60° = \sqrt{3}$

d. $\tan(-45°)$

d
$-\tan 45° = -1$

e. $\sin 135°$

e
$\sin 45° = \frac{1}{\sqrt{2}}$

$\sin(x) = \sin(180 - x)$
$\cos(x) = \cos(360 - x)$
sin, cos repeat every 360° but tan every 180°

Exercise 10B – Y1 Page 209 — Solution 10

Use a reference angle of 30°, 45° or 60° to find each exact value.

(a) $\sin 135°$

a
$\sin 45° = \frac{\sqrt{2}}{2}$

(b) $\sin(-60°)$

b
$-\sin 60° = -\frac{\sqrt{3}}{2}$

(c) $\sin 330°$

c
$-\sin 30° = -\frac{1}{2}$

(d) $\sin 420°$

d
$\sin 60° = \frac{\sqrt{3}}{2}$

(e) $\sin(-300°)$

e
$\sin 60° = \frac{\sqrt{3}}{2}$

(f) $\cos 120°$

f
$-\cos 60° = -\frac{1}{2}$

(g) $\cos 300°$

g
$\cos 60° = \frac{1}{2}$

(h) $\cos 225°$

h
$-\cos 45° = -\frac{\sqrt{2}}{2}$

(i) $\cos(-210°)$

i
$-\cos 30° = -\frac{\sqrt{3}}{2}$

(j) $\cos 495°$

j
$-\cos 45° = -\frac{\sqrt{2}}{2}$

(k) $\tan 135°$

k
$-\tan 45° = -1$

(l) $\tan(-225°)$

l
$-\tan 45° = -1$

(m) $\tan 210°$

m
$\tan 30° = \frac{\sqrt{3}}{3}$

(n) $\tan 300°$

n
$-\tan 60° = -\sqrt{3}$

(o) $\tan(-120°)$

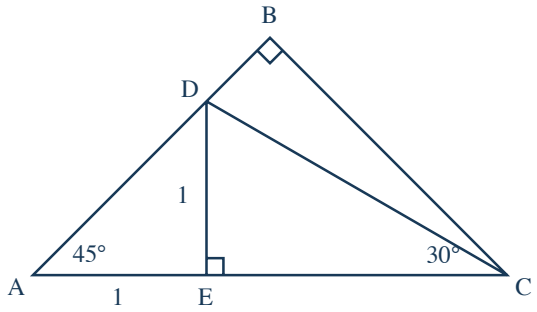
o
$\tan 60° = \sqrt{3}$

Exercise 10A/B – Y1 Page 207, 209

Solution 11

Challenge: exact values for 15°

ABC is an isosceles right-angled triangle, with the right angle at B. D lies on AB and E lies on AC. AE = DE = 1, DE is perpendicular to AC, and angle ACD = 30°.



(a)

Find the exact lengths CE, DC, BC and DB.

a

$$CE = \frac{1}{\tan 30^\circ} = \sqrt{3}, \quad DC = \sqrt{1+3} = 2$$

$$BC = \frac{AC}{\sqrt{2}} = \frac{1+\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{2}$$

$$DB = AB - AD = \frac{\sqrt{6}-\sqrt{2}}{2}$$

(b)

Find angle BCD.

b $\angle BCD = 45^\circ - 30^\circ = 15^\circ$

(c)

Find exact values of $\sin 15^\circ$ and $\cos 15^\circ$.

c

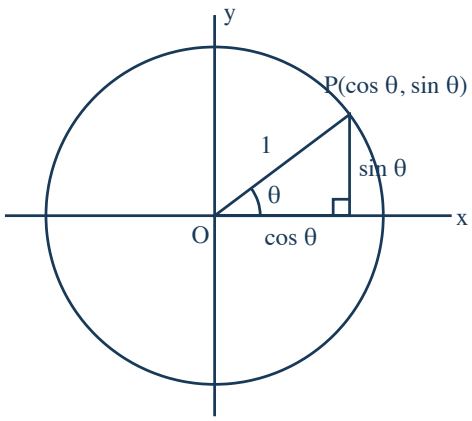
$$\sin 15^\circ = \frac{DB}{DC} = \frac{\sqrt{6}-\sqrt{2}}{4}$$

$$\cos 15^\circ = \frac{BC}{DC} = \frac{\sqrt{6}+\sqrt{2}}{4}$$

Trigonometric Identities

Solution 12

Two trigonometric identities



Here $\sin^2\theta$ means $(\sin\theta)^2$, not $\sin(\theta^2)$.

1.

Opposite divided by adjacent

1 $$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

This identity holds whenever cosine is nonzero.

2.

Pythagoras on the unit circle

2 $$\cos^2\theta + \sin^2\theta = 1$$

The identity holds for every real angle.

Application of identities #1: Proofs

Solution 13

Prove the identity (where the expressions are defined).

$1 - \tan\theta \sin\theta \cos\theta \equiv \cos^2\theta$

Method

Start with one side and transform it into the other.
Replace tangent by sine divided by cosine.

$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

Proof

$$\begin{aligned} 1 - \tan\theta \sin\theta \cos\theta &= 1 - \frac{\sin\theta}{\cos\theta} \sin\theta \cos\theta \\ &= 1 - \sin^2\theta \\ &= \cos^2\theta. \end{aligned}$$

The symbol $\equiv$ denotes an identity: equality for every angle in the common domain. Here cosine must be non-zero because tangent appears.

More Examples

Solution 14

Prove the identity where both sides are defined.

$\tan\theta + \dfrac{1}{\tan\theta} \equiv \dfrac{1}{\sin\theta \cos\theta}$

$$\begin{aligned} \tan\theta + \frac{1}{\tan\theta} &= \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} \\ &= \frac{\sin^2\theta + \cos^2\theta}{\sin\theta \cos\theta} \\ &= \frac{1}{\sin\theta \cos\theta}. \end{aligned}$$

Combine fractions using a common denominator.

Simplify.

$5 - 5\sin^2\theta$

$5(1 - \sin^2\theta) = 5\cos^2\theta$

Look for $1 - \sin^2\theta = \cos^2\theta$ or $1 - \cos^2\theta = \sin^2\theta$.

Test Your Understanding — Solution 15

Prove each identity where its expressions are defined.

$\tan^2\theta \equiv \dfrac{1}{\cos^2\theta} - 1$

$\tan^2\theta = \dfrac{\sin^2\theta}{\cos^2\theta}$

$= \dfrac{1-\cos^2\theta}{\cos^2\theta} = \dfrac{1}{\cos^2\theta} - 1$

For $0 < x < 90°$, prove:

$\dfrac{\tan x \cos x}{\sqrt{1-\cos^2 x}} = 1$

$\dfrac{\sin x}{\sqrt{\sin^2 x}} = \dfrac{\sin x}{|\sin x|} = 1$

Sine is positive in this interval.

$\dfrac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

$\dfrac{(\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$

$= \dfrac{\cos^2\theta - \sin^2\theta}{\cos^2\theta} = 1 - \tan^2\theta$

Exercise 10C – Y1 Page 211-212 — Solution 16

10.

Eliminate θ to give an equation relating x and y in each case.

(a)

$x = \sin\theta, \quad y = \cos\theta$

a

$x^2 + y^2 = \sin^2\theta + \cos^2\theta = 1$

(b)

$x = \sin\theta, \quad y = 2\cos\theta$

b

$x^2 + \dfrac{y^2}{4} = 1$

(c)

$x = \sin\theta, \quad y = \cos^2\theta$

c

$x^2 + y = 1$

(d)

$x = \sin\theta, \quad y = \tan\theta$

d

$\cos\theta = \dfrac{x}{y} \quad (y \neq 0)$

$x^2y^2 + x^2 = y^2$

The resulting equation also holds at x = y = 0.

(e)

$x = \sin\theta + \cos\theta$

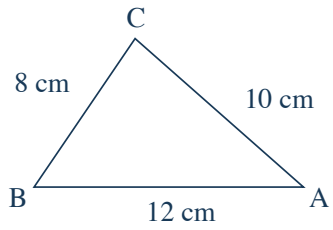
$y = \cos\theta - \sin\theta$

e

$x^2 + y^2 = 2\sin^2\theta + 2\cos^2\theta = 2$

Exercise 10C – Y1 Page 211-212 Solution 17

11. Triangle ABC has $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.



a. Show that $\cos B = \frac{9}{16}$. (3 marks)

11a.

Using the cosine rule:

$\cos B = \frac{a^2+c^2-b^2}{2ac}$

$= \frac{8^2+12^2-10^2}{2\times 8\times 12}$

$= \frac{64+144-100}{192} = \frac{108}{192} = \frac{9}{16}$.

b. Hence find the exact value of $\sin B$. (2 marks)

11b.

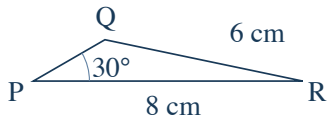
$\sin^2\theta + \cos^2\theta = 1$.

$\sin^2 B + (\frac{9}{16})^2 = 1$.

$\sin^2 B = 1 - \frac{81}{256} = \frac{175}{256}$.

$\sin B = \sqrt{\frac{175}{256}} = \frac{5\sqrt{7}}{16}$.

12. Triangle PQR has $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and $\angle QPR = 30°$.



a. Show that $\sin Q = \frac{2}{3}$. (3 marks)

12a.

Using the sine rule:

$\frac{\sin Q}{q} = \frac{\sin P}{p}$.

$\frac{\sin Q}{8} = \frac{\sin 30°}{6}$.

$\sin Q = \frac{8\sin 30°}{6} = \frac{8\times\frac{1}{2}}{6} = \frac{2}{3}$.

b. Given that Q is obtuse, find the exact value of $\cos Q$. (2 marks)

12b.

$\sin^2 Q + \cos^2 Q \equiv 1$.

$(\frac{2}{3})^2 + \cos^2 Q = 1$.

$\cos^2 Q = 1 - \frac{4}{9} = \frac{5}{9}$.

Q is obtuse (quadrant II), where cosine is negative.

$\cos Q = -\frac{\sqrt{5}}{3}$.

Extension Solution 18

[MAT 2008 1C] For which angles in $0 \le \theta < 2\pi$ are these simultaneous equations solvable?

$x\cos\theta - y\sin\theta = 2$

$x\sin\theta + y\cos\theta = 1$

A: all angles; B: all except one; C: all except two; D: all except three.

Answer

Let $s = \sin\theta$ and $c = \cos\theta$. Multiply the first equation by s and the second by c, then subtract:

$(s^2 + c^2)y = c - 2s$

$y = c - 2s$

Similarly, eliminate y:

$x = 2c + s$

Both are defined for every θ. Answer A.

No division by a potentially zero sine or cosine is required. Dividing by one could incorrectly exclude valid angles.

Solving Trigonometric Equations

Solution 19

Solve $\sin\theta = \frac{1}{2}$ for $0 \leq \theta \leq 360°$.

$\theta = \sin^{-1}(1/2) = 30°$

$\theta = 180° - 30° = 150°$

An inverse trigonometric function on a calculator returns only the principal value.

$\sin\theta = \sin(180° - \theta)$

$\cos\theta = \cos(360° - \theta)$

Sine and cosine repeat every $360°$; tangent repeats every $180°$.

Solve $5\tan\theta = 10$ for $-180° \leq \theta < 180°$.

$\tan\theta = 2$

$\theta = 63.4°$ or $-116.6°$

Tangent repeats every $180°$. Always check the stated interval.

Slightly Harder Ones...

Solution 20

Solve $\sin\theta = -\frac{1}{2}$ for $0 \leq \theta \leq 360°$.

$\sin^{-1}(-1/2) = -30°$

$180° - (-30°) = 210°$

$-30° + 360° = 330°$

The principal value $-30°$ is outside the interval. The required answers are $210°$ and $330°$.

Solve $\sin\theta = \sqrt{3}\cos\theta$ for $0 \leq \theta \leq 360°$.

Express the equation in one trigonometric function.

Cosine cannot be zero in the original equation, so division by cosine is valid.

$\tan\theta = \sqrt{3}$

$\theta = 60°$ or $240°$

Test Your Understanding — Solution 21

Solve $2\cos\theta = \sqrt{3}$ for $0 \le \theta \le 360°$.

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30° \quad \text{or} \quad 330°$$

Solve $\sqrt{3}\sin\theta = \cos\theta$ for $-180° \le \theta \le 180°$.

$$\tan\theta = \frac{1}{\sqrt{3}}$$

$$\theta = 30° \quad \text{or} \quad -150°$$

Exercise 10D – Y1 215-216 — Solution 22

7.

Solve $2\cos x = 3\sin x$ for $-180° \le x \le 180°$.

Student A obtains $\tan x = 3/2$ and $x = 56.3°$ or $-123.7°$.

Student B squares both sides and obtains $\sin x = \pm\sqrt{(4/13)}$, giving $x = \pm 33.7°$ or $\pm 146.3°$.

(a)

Identify Student A's mistake.

7(a)

$$\tan x = \frac{2}{3}, \text{ not } \frac{3}{2}.$$

(b)

Identify Student B's mistake and its effect.

7(b)

Squaring loses the sign relationship between sine and cosine. It introduces two extra solutions, $-33.7°$ and $146.3°$.

(c)

Give the correct answers.

7(c)

$$x = 33.7° \quad \text{or} \quad -146.3°$$

9.

Solve $\tan^2\theta = 9$ for $0 < \theta < 360°$, to 1 decimal place.

9

$$\tan\theta = \pm 3$$

$$\theta = 71.6°, 108.4°, 251.6°, 288.4°$$

Exercise 10D – Y1 215-216

Solution 23

10(a)

Show that $4\sin^2 x - 3\cos^2 x = 2$ becomes $7\sin^2 x = 5$.

10(a)

$4\sin^2 x - 3(1 - \sin^2 x) = 2$

$7\sin^2 x = 5$

10(b)

Hence solve for $0 \leq x \leq 360^\circ$, giving answers to 1 decimal place.

10(b)

$\sin x = \pm\sqrt{5/7}$

$x = 57.7^\circ, 122.3^\circ, 237.7^\circ, 302.3^\circ$

11(a)

Show that $2\sin^2 x + 5\cos^2 x = 1$ becomes $3\sin^2 x = 4$.

11(a)

$2\sin^2 x + 5(1 - \sin^2 x) = 1$

$3\sin^2 x = 4$

11(b)

Explain why there are no real solutions.

11(b)

$\sin^2 x = \frac{4}{3} > 1$

For every real x, $0 \leq \sin^2 x \leq 1$. Hence no real solution exists.

Harder Equations

Solution 24

Harder questions replace the angle θ with a linear expression.

Solve $\cos 3x = -\frac{1}{2}$ in the interval $0 \leq x \leq 360^\circ$.

STEP 1. Adjust the range for the expression inside the cosine.

STEP 2. Immediately after applying the inverse trigonometric function, find every solution up to the end of the interval—before dividing by 3.

STEP 3. Apply the final manipulation to every value.

1.

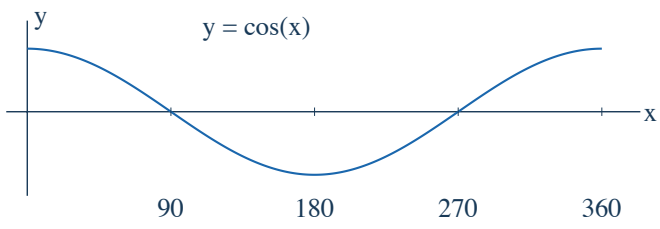
$0 \leq 3x \leq 1080^\circ.$

$3x = \cos^{-1}(-\frac{1}{2}) = 120^\circ.$

$3x = 120^\circ, 240^\circ, 480^\circ, 600^\circ, 840^\circ, 960^\circ.$

$x = 40^\circ, 80^\circ, 160^\circ, 200^\circ, 280^\circ, 320^\circ.$

In general, sine, cosine and tangent each give a pair of values per 360°, except when $\cos\theta = \pm 1$ or $\sin\theta = \pm 1$.
Find the first pair using $\sin(180^\circ - \theta)$ or $\cos(360^\circ - \theta)$, then keep adding 360° to generate new pairs.



Further Examples

Solution 25

Solve $\sin(2x + 30°) = \frac{1}{\sqrt{2}}$ for $0 \leq x \leq 360°$.

$30° \leq 2x + 30° \leq 750°$

$2x + 30° = 45°, 135°, 405°, 495°$

$2x = 15°, 105°, 375°, 465°$

$x = 7.5°, 52.5°, 187.5°, 232.5°$

Transform both endpoints of the interval before finding all the angles.

Solve $\sin x = 2\cos x$ for $0 \leq x < 300°$.

Cosine is nonzero for any solution, so divide by cosine.

$\tan x = 2$

$x = 63.43°$ or $243.43°$

Test Your Understanding

Solution 26

Solve $\cos(3x - 10°) = -0.4$ for $0 \leq x < 180°$. Show each step and give answers to 1 decimal place. (7 marks)

Adjust the interval

$-10° \leq 3x - 10° < 530°$

Find every intermediate angle

$\alpha = \cos^{-1}(-0.4) = 113.578\ldots°$

$3x - 10° = \alpha,\ 360° - \alpha,\ 360° + \alpha$

Add $10°$ and divide each angle by 3

$x = 41.2°,\ 85.5°,\ 161.2°$

Exercise 10E – Y1 p219 — Solution 27

4.

Solve for $0 \leq x \leq 180^\circ$.

(a)

$\sin(x + 20^\circ) = \frac{1}{2}$

4(a)

$20^\circ \leq x + 20^\circ \leq 200^\circ$

$x + 20^\circ = 30^\circ, 150^\circ$

$x = 10^\circ, 130^\circ$

(b)

Give answers to 1 decimal place.

$\cos 2x = -0.8$

4(b)

$0 \leq 2x \leq 360^\circ$

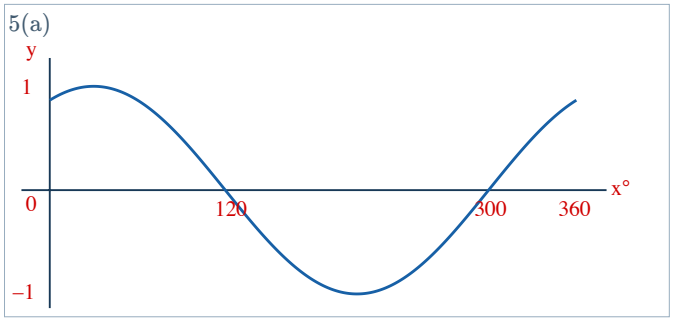
$2x = 143.13\ldots^\circ, 216.87\ldots^\circ$

$x = 71.6^\circ, 108.4^\circ$

5(a)

Sketch $y = \sin(x + 60^\circ)$ for $0 \leq x \leq 360^\circ$.

5(a)



5(b)

Give the exact axis intercepts.

5(b) $(0, \sqrt{3}/2), \quad (120^\circ, 0), \quad (300^\circ, 0)$

5(c)

Solve $\sin(x + 60^\circ) = 0.55$ on the same interval, to 1 decimal place.

5(c)

$60^\circ \leq x + 60^\circ \leq 420^\circ$

$x + 60^\circ = 146.63\ldots^\circ, 393.36\ldots^\circ$

$x = 86.6^\circ, 333.4^\circ$

Exercise 10E – Y1 p219 — Solution 28

6(a)

Given $4\sin x = 3\cos x$, find $\tan x$.

6(a)

$\tan x = \frac{3}{4}$

6(b)

Solve $4\sin 2\theta = 3\cos 2\theta$ for $0 \leq \theta \leq 360^\circ$, to 1 decimal place.

6(b)

$\tan 2\theta = \frac{3}{4}, \quad 0 \leq 2\theta \leq 720^\circ$

$2\theta = 36.87\ldots^\circ + 180^\circ n, \quad n = 0, 1, 2, 3$

$\theta = 18.4^\circ, 108.4^\circ, 198.4^\circ, 288.4^\circ$

7

The equation $\tan(kx) = -1/\sqrt{3}$, where $k > 0$, has a solution at $x = 60^\circ$.

(a)

Find a possible k.

7(a) $60k = 150 \quad \Rightarrow \quad k = 2.5$

(b)

Is this the only possible value? Justify your answer.

7(b)

$60k = -30 + 180n$

$k = -\frac{1}{2} + 3n, \quad n = 1, 2, 3, \ldots$

There are infinitely many positive values: 2.5, 5.5, 8.5, ...

The period of tan(kx) is $180^\circ/k$, so it decreases as k increases.

Quadratics in sin/cos/tan

Solution 29

Solve $5\sin^2 x + 3\sin x - 2 = 0$ for $0 \le x \le 360°$.

Method 1: substitute

$y = \sin x$

$5y^2 + 3y - 2 = (5y - 2)(y + 1) = 0$

$y = \frac{2}{5} \quad \text{or} \quad y = -1$

$\sin x = \frac{2}{5} \quad \text{or} \quad \sin x = -1$

$x = 23.6°,\ 156.4°,\ 270°$

Method 2: factorise directly

$(5\sin x - 2)(\sin x + 1) = 0$

$\sin x = \frac{2}{5} \quad \text{or} \quad \sin x = -1$

$x = 23.6°,\ 156.4°,\ 270°$

"Quadratic in sine" means that replacing $\sin x$ with a single variable produces a quadratic. Similarly $x - 2\sqrt{x} + 1 = 0$ is quadratic in $\sqrt{x}$.

More Examples

Solution 30

Solve $\tan^2\theta = 4$ for $0 \le \theta \le 360°$.

$\tan\theta = 2 \quad \text{or} \quad \tan\theta = -2$

$\theta = 63.4°,\ 116.6°,\ 243.4°,\ 296.6°$

Include both signs of the square root. Tangent repeats every 180°; discard angles outside the stated interval.

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ for $-180° \le x \le 180°$.

2

$2(1 - \sin^2 x) + \sin x = 3\sin^2 x$

$5\sin^2 x - \sin x - 2 = 0$

$\sin x = \frac{1 \pm \sqrt{41}}{10}$

$x \approx -147.3°,\ -32.7°,\ 47.8°,\ 132.2°$

Use the identity to express the equation in one trigonometric function, then use the quadratic formula.

Test Your Understanding — Solution 31

a. Show that $5\sin x = 1 + 2\cos^2 x$ can be written as $2\sin^2 x + 5\sin x - 3 = 0$.

a

$5\sin x = 1 + 2(1 - \sin^2 x)$

$5\sin x = 3 - 2\sin^2 x$

$2\sin^2 x + 5\sin x - 3 = 0.$

b. Solve $2\sin^2 x + 5\sin x - 3 = 0$ for $0 \leq x < 360°$.

b

$(2\sin x - 1)(\sin x + 3) = 0.$

$\sin x = \frac{1}{2}$ or $\sin x = -3.$

The second is impossible. Thus $x = 30°, 150°.$

Exercise 10F – Y1 P221-222 — Solution 32

5.

Find all solutions of $8\sin^2 x + 6\cos x - 9 = 0$ for $0 \leq x \leq 360°$, to 1 decimal place. (6 marks)

5: rearrange

$8(1 - \cos^2 x) + 6\cos x - 9 = 0$

$8\cos^2 x - 6\cos x + 1 = 0$

$(4\cos x - 1)(2\cos x - 1) = 0$

5: solve

$\cos x = \frac{1}{4}$ or $\frac{1}{2}$

$x = 60.0°, 75.5°, 284.5°, 300.0°$

6.

Find all solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ for $0 \leq x \leq 360°$, to 1 decimal place.

6: rearrange

$2\sin^2 x + 2 = 7(1 - \sin^2 x)$

$9\sin^2 x = 5$

$\sin x = \pm\frac{\sqrt{5}}{3}$

6: solve

$x = 48.2°, 131.8°, 228.2°, 311.8°$

Exercise 10F – Y1 P221-222 Solution 33

7.

Show that $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

7

$(2\cos x - 3)(\cos x + 2) = 0$

$\cos x = \frac{3}{2}$ or -2

Both are outside $[-1, 1]$, so there are no real solutions.

8(a)

Show that $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

8(a)

$1 - \sin^2 x = 2 - \sin x$

$\sin^2 x - \sin x + 1 = 0$

8(b)

Hence show that the original equation has no solutions. (3 marks)

8(b)

$b^2 - 4ac = (-1)^2 - 4(1)(1) = -3 < 0$

The quadratic has no real roots, so no real value of sin x can satisfy the equation.

Exercise 10F – Y1 P221-222 Solution 34

9.

$\tan^2 x - 2\tan x - 4 = 0$

(a)

Show that $\tan x = p \pm \sqrt{q}$, finding p and q.

a

$\tan x = \frac{2 \pm \sqrt{4 + 16}}{2} = 1 \pm \sqrt{5}$

$p = 1, \quad q = 5$

(b)

Hence solve the equation for $0 \leq x \leq 540^\circ$.

b: angles

$\tan x = 1 + \sqrt{5} \Rightarrow x = 72.8^\circ + 180^\circ n$

$\tan x = 1 - \sqrt{5} \Rightarrow x = -51.0^\circ + 180^\circ n$

b: solutions

$x = 72.8^\circ, 129.0^\circ, 252.8^\circ,$

$309.0^\circ, 432.8^\circ, 489.0^\circ$

1.

[MAT 2010 1C]

How many solutions are there for $0 \leq x < 360°$?

$\sin^2 x + 3\sin x \cos x + 2\cos^2 x = 0$

1

Cosine cannot be zero at a solution. Divide by its square:

$\tan^2 x + 3\tan x + 2 = 0$

$(\tan x + 1)(\tan x + 2) = 0$

Each of tan x $= -1$ and tan x $= -2$ has two solutions in the interval. Hence there are four solutions.

2.

[MAT 2014 1E]

Find the largest value, for real x, of

$(4\sin^2 x + 4\cos x + 1)^2$

2

$4\sin^2 x + 4\cos x + 1 = 6 - (2\cos x - 1)^2$

Since $-1 \leq \cos x \leq 1$, the inner expression lies between -3 and 6. Its square is therefore at most 36.

$\cos x = \frac{1}{2} \Rightarrow (4\sin^2 x + 4\cos x + 1)^2 = 36$